

IOT BASED REMOTE PATIENT VITAL MONITORING SYSTEM

ABSTRACT

Access to timely healthcare remains a significant challenge in rural, remote and disaster-prone regions where medical infrastructure and specialist support are limited. To address this issue, this project proposes the development of a portable IoT-based Patient Vitals Monitoring System that integrates multiple sensors into a single compact device for real-time patient health assessment. The system is designed to measure critical vital parameters such as heart rate, blood oxygen saturation (SpO₂), body temperature, blood pressure, etc enabling comprehensive monitoring of a patient's condition. The collected physiological data are processed and transmitted through Internet of Things (IoT) technology to a secure cloud platform, allowing healthcare professionals to remotely access patient information in real time.

This capability enables doctors and medical experts to monitor patients, provide preliminary diagnoses, and make informed clinical decisions without being physically present at the patient's location. The system is particularly beneficial for under-served communities, emergency response scenarios, rural healthcare centers, mobile medical units, and home-based patient care. By bridging the gap between patients and healthcare providers, the proposed IoT-based Medical Vitals Monitoring System aims to improve healthcare accessibility, facilitate early detection of critical conditions, reduce response times during emergencies, and ultimately enhance patient outcomes. This project represents a cost-effective and scalable approach toward strengthening remote healthcare services and advancing telemedicine applications in modern healthcare systems.

TABLE OF CONTENTS

CHAPTER NO	TITLE	PAGE NO
	ABSTRACT	v
	LIST OF TABLES	vi
	LIST OF FIGURES	vii
1	INTRODUCTION	1
	1.1 INTRODUCTION	1
	1.2 BACKGROUND OF THE STUDY	2
	1.3 PROBLEM STATEMENT	3
	1.4 OBJECTIVES OF THE PROJECT	3
	1.5 SCOPE OF THE PROJECT	4
2	LITERATURE SURVEY	6
3	PROPOSED SYSTEM	9
	3.1 OVERVIEW OF THE PROPOSED SYSTEM	9
	3.2 SYSTEM COMPONENTS	10
	3.3 WORKING OF PROPOSED SYSTEM	10
	3.4 FEATURES OF THE PROPOSED SYSTEM	12
	3.5 ADVANTAGES OF PROPOSED SYSTEM	12
4	HARDWARE AND SOFTWARE DESCRIPTION	13
	4.1 HARDWARE COMPONENTS	13
	4.2 SOFTWARE DESCRIPTION	17
		5
		SYSTEM DESIGN
		19
	5.1 CIRCUIT DIAGRAM	19
	5.2 WORKING PRINCIPLE	19
	5.3 ALGORITHM	20
	5.4 FLOWCHART	21
6	SYSTEM IMPLEMENTATION AND RESULTS	22
	6.1 IMPLEMENTATION	22
	6.2 EXPERIMENTAL SETUP	22
	6.3 TESTING METHODOLOGY	23
	6.4 TEST CASES AND RESULTS	24

	6.5 RESULT ANALYSIS	24	
	6.6 PROTOTYPE MODEL	25	
7	CONCLUSION	27	
	REFERENCES		29

LIST OF FIGURES

Fig no	Figures	Page no
3.1	Block Diagram of Proposed System	11
4.1	ESP32 Development Board	13
4.2	MLX90614 Infrared Temperature Sensor	14
4.3	AD8232 ECG Heart Rate Sensor Module	15
4.4	MAX30102 Pulse Oximeter Sensor	16
4.5	Breadboard and Connecting Wires	17
5.1	IoT-Based Remote Patient Vital Monitoring System.	19
5.2	Flowchart of the Proposed System	21
6.1	Dashboard of the IoT-Based Remote Patient Vital Monitoring System	25
6.2	Dashboard of the IoT-Based Remote Patient Vital Monitoring System	26
6.3	Prototype model for IoT-Based Remote Monitoring System	26

CHAPTER 1 INTRODUCTION

1.1 Introduction

The rapid advancement of information and communication technologies has transformed the healthcare sector by introducing smart and connected medical systems. Among these advancements, the Internet of Things (IoT) has emerged as a revolutionary technology that enables real-time monitoring, data collection, and remote communication between medical devices and healthcare professionals. IoT-based healthcare systems improve the quality of patient care by allowing continuous monitoring of vital health parameters without requiring patients to remain in hospitals. Traditional healthcare systems primarily rely on periodic hospital visits for monitoring patient health. This approach is often inconvenient, expensive, and time-consuming, particularly for elderly patients, individuals with chronic diseases, and those living in rural or remote areas where healthcare facilities are limited. During emergencies or infectious disease outbreaks, frequent hospital visits may also increase health risks and place additional burdens on medical resources. Therefore, there is a growing demand for portable and intelligent healthcare systems that provide continuous monitoring while reducing dependency on hospital-based care.

The proposed IoT-Based Remote Patient Vital Monitoring System addresses these challenges by integrating multiple biomedical sensors into a compact and portable device capable of measuring essential physiological parameters such as heart rate, blood oxygen saturation (SpO_2), body temperature, and blood pressure. The measured data are processed by a microcontroller and transmitted securely through IoT technology to a cloud platform where healthcare professionals can access the patient's health information in real time. This system enables doctors, nurses, and caregivers to remotely monitor patients from any location using internet-connected devices such as smartphones, tablets, or computers. Continuous monitoring helps in the early detection of abnormal health conditions, allowing timely medical intervention and reducing the risk of serious complications. The portability, affordability, and ease of operation make the proposed system suitable for home healthcare, rural clinics, elderly care, ambulances, and emergency medical services. Overall, the project aims to enhance healthcare accessibility, improve patient safety, reduce hospitalization costs, and support the growing demand for digital healthcare solutions through the effective integration of IoT technology and biomedical sensing.

1.2 Background of the Study

Healthcare systems worldwide are facing increasing challenges due to the rising number of patients suffering from chronic diseases such as hypertension, diabetes, cardiovascular disorders, and respiratory illnesses. Continuous monitoring of patients is essential for effective diagnosis, treatment, and disease management. However, conventional monitoring methods often require patients to remain under hospital supervision or make frequent visits to healthcare facilities, which may not always be practical or affordable. The development of IoT technology has opened new opportunities for remote healthcare applications by connecting medical devices through the internet. IoT-based healthcare systems collect physiological data using sensors, process the information using embedded systems, and transmit it to cloud servers for remote access by healthcare professionals. This approach enables continuous observation of patient health regardless of geographical location.

Recent advancements in microcontrollers, wireless communication technologies, cloud computing, and low-cost biomedical sensors have made remote patient monitoring systems more reliable, portable, and affordable. Sensors capable of measuring heart rate, blood oxygen level, body temperature, and blood pressure can now be integrated into a single compact device, enabling comprehensive health assessment. The increasing adoption of telemedicine and digital healthcare services has further accelerated the demand for IoT-enabled patient monitoring systems. Such systems not only improve healthcare accessibility but also reduce hospital workload, minimize unnecessary hospital admissions, and enable early diagnosis through continuous monitoring of patient health trends. This study focuses on designing and developing a portable IoT-based patient monitoring system capable of measuring multiple vital parameters simultaneously and transmitting them securely for real-time remote monitoring. This study focuses on designing and developing a portable IoT-based patient monitoring system capable of measuring multiple vital parameters simultaneously and transmitting them securely for real-time remote monitoring.

1.3 Problem Statement

Many patients suffering from chronic illnesses, cardiovascular diseases, respiratory disorders, or age-related health conditions require continuous monitoring of their vital signs. Traditional patient monitoring methods are largely

hospitalcentered and depend on regular clinical visits or prolonged hospitalization, which can be costly, inconvenient, and inaccessible for many individuals. Patients living in rural and remote areas often face difficulties in accessing specialized healthcare facilities due to inadequate medical infrastructure, long travel distances, and shortage of healthcare professionals. Delayed diagnosis resulting from infrequent monitoring may lead to severe medical complications and increased mortality.

Existing healthcare systems generally monitor only one or two physiological parameters at a time or require expensive medical equipment that is unsuitable for home-based healthcare. Moreover, continuous manual monitoring increases the workload on healthcare staff and limits the efficient utilization of hospital resources. Therefore, there is a need for a portable, low-cost, reliable, and IoT-enabled patient monitoring system capable of continuously measuring multiple vital parameters, securely transmitting patient data over the internet, and enabling healthcare professionals to remotely monitor patient health in real time. Such a system can improve early diagnosis, reduce healthcare costs, and provide better medical assistance to patients regardless of their location.

1.4 Objectives of the Project

The main objective of this project is to develop a portable IoT-Based Remote Patient Vital Monitoring System capable of continuously monitoring essential physiological parameters and transmitting them securely for remote healthcare applications.

Specific Objectives

- To design and develop a compact patient monitoring device using IoT technology.
- To measure heart rate accurately using a biomedical pulse sensor.
- To monitor blood oxygen saturation (SpO_2) continuously using an oxygen saturation sensor.
- To measure body temperature using a digital temperature sensor.
- To monitor blood pressure for early detection of hypertension and cardiovascular abnormalities.

- To integrate multiple sensors into a single embedded system.
- To transmit patient health data securely through an IoT platform for real-time remote monitoring.
- To enable doctors and caregivers to access patient information using smartphones or computers.
- To improve early diagnosis through continuous monitoring of patient vital signs.
- To reduce hospital visits and support home-based healthcare services.
- To develop a portable, affordable, and user-friendly healthcare monitoring system.

1.5 Scope of the Project

The scope of this project includes the design, implementation, and testing of a portable IoT-enabled patient monitoring device capable of measuring four essential physiological parameters: heart rate, blood oxygen saturation (SpO₂), body temperature, and blood pressure. The system integrates biomedical sensors with a microcontroller for data acquisition and processing. The collected patient data are transmitted through the Internet of Things to a cloud platform, allowing authorized healthcare professionals to monitor patient health remotely in real time. The system is intended for use in homes, hospitals, rural healthcare centers, ambulances, elderly care facilities, and disaster relief operations where continuous monitoring is essential.

The proposed system focuses on providing:

- Continuous real-time monitoring of vital signs.
- Remote access to patient health data through IoT.
- Early detection of abnormal physiological conditions.
- Reduced hospital visits and healthcare costs.
- Support for telemedicine and home healthcare.
- A portable and cost-effective healthcare solution.

Although the system provides continuous monitoring and remote accessibility, it is designed as a health monitoring and decision-support tool rather than a

replacement for professional medical diagnosis or emergency treatment. Future enhancements may include artificial intelligence for disease prediction, ECG integration, wearable sensor support, GPS-based patient tracking, emergency alert notifications, and electronic health record (EHR) integration to create a more comprehensive smart healthcare system.

CHAPTER 2 LITERATURE REVIEW

“Smart Healthcare Monitoring Using Internet of Things” Authors: M. S. Hossain, G. Muhammad, and A. Alamri | Year of Publishing: 2015

This paper presents an IoT-based smart healthcare monitoring system that enables continuous monitoring of patient vital signs using wearable sensors and cloud technology. The system collects physiological data such as heart rate, body temperature, and ECG, and transmits it to healthcare professionals through the Internet for remote diagnosis. The study highlights that IoT improves healthcare accessibility, reduces hospitalization costs, and supports early detection of medical emergencies. However, the system mainly focuses on cloud-based monitoring and does not provide an integrated low-cost portable solution for multiple health parameters.

“The Internet of Things for Health Care: A Comprehensive Survey” Authors: S. M. Riazul Islam, D. Kwak, M. H. Kabir, M. Hossain, and K. S. Kwak | Year of Publishing: 2015

This survey provides a comprehensive overview of IoT applications in healthcare. It discusses the architecture of IoT healthcare systems, wearable sensors, wireless communication technologies, cloud computing, and data security. The authors conclude that IoT-based healthcare systems enable continuous patient monitoring and improve healthcare services through remote access to medical information. However, the survey identifies challenges such as data privacy, interoperability, and network reliability that need further improvement.

“IoT-Based Wireless Patient Monitor Using ESP32 Microcontroller” Authors: K. K. Mujeeb Rahman, Nasor N. Mohamed, Rayan Zidan, Ibrahim Alsarraj, and Beshar Hasan | Year of Publishing: 2023

This research proposes an ESP32-based wireless patient monitoring system capable of measuring ECG, heart rate, and blood oxygen saturation (SpO₂). The collected data are transmitted to a Firebase cloud database using the ESP32's builtin Wi-Fi module, allowing doctors to monitor patients remotely in real time. The system is particularly useful for elderly patients and individuals living in remote areas. However, the work does not include additional parameters such as body temperature and blood pressure monitoring, limiting its overall healthcare assessment capability.

“Patient Health Monitoring System Using IoT” Authors: P. Karthik et al. | Year of Publishing: 2021

This paper presents an IoT-enabled patient monitoring system that measures heart rate, ECG, body temperature, blood pressure, and SpO₂ using biomedical sensors connected to an ESP32 microcontroller. The sensor data are uploaded to the cloud through Wi-Fi and displayed on a mobile application for remote monitoring. The system also generates alerts when any measured parameter exceeds predefined threshold values. Although the system provides comprehensive monitoring, it requires further optimization in terms of portability, power consumption, and data security.

“A Portable IoT-Based Health Monitoring Framework Using ESP32 for Isolated and Home-Based Patient Care” Authors: Hemaxi Vyas, Hitasvi Shukla, and Mahesh N. Jivani |Year of Publishing: 2025

This paper introduces a portable IoT-based health monitoring framework using the ESP32 microcontroller for home-based patient care. The system measures heart rate, blood oxygen saturation (SpO₂), and body temperature using biomedical sensors and transmits the data to a cloud platform for real-time monitoring. The authors emphasize the portability, affordability, and ease of deployment of the system, making it suitable for home healthcare and telemedicine applications. However, the proposed framework does not include ECG and blood pressure monitoring, which are important for comprehensive cardiovascular assessment.

“Real-Time Patient's Vital Sign Monitoring Using ESP32”Authors: Nur Farahdinna Anuar, Mohammad Hafidz Rahmat, and Mohd Hafifi Mohd Supir | Year of Publishing: 2025

This study presents a real-time patient monitoring system based on the ESP32 microcontroller. The system continuously measures heart rate, SpO₂, and body temperature and uploads the collected data to the ThingSpeak cloud platform. It also integrates a Telegram notification service to send alerts when abnormal readings are detected. The proposed system demonstrates reliable cloud communication and effective remote monitoring. However, it does not monitor blood pressure or ECG signals, limiting its use for complete cardiovascular health assessment.

CHAPTER 3 PROPOSED SYSTEM

3.1 Overview of the Proposed System

The IoT-Based Remote Patient Vital Monitoring System is designed to provide a portable, reliable, and cost-effective solution for continuously monitoring a patient's vital health parameters. The system integrates multiple biomedical sensors with an embedded microcontroller and Internet of Things (IoT) technology to collect, process, and transmit patient health data in real time. The primary objective of the system is to assist healthcare professionals in remotely monitoring patients without requiring them to remain in hospitals.

The proposed system continuously measures four essential physiological parameters:

- Heart Rate (BPM)
- Blood Oxygen Saturation (SpO₂)
- Body Temperature
- Blood Pressure

Each sensor continuously acquires physiological data from the patient and sends it to the microcontroller for processing. After processing, the information is transmitted through a wireless communication module to a cloud server using Internet connectivity. Doctors, nurses, and caregivers can access the patient's health information remotely through smartphones, tablets, or computers. Unlike

conventional monitoring systems, which require patients to visit hospitals frequently, the proposed system enables continuous monitoring regardless of the patient's location. This makes it particularly useful for elderly patients, individuals with chronic diseases, rural healthcare centers, home-based healthcare services, ambulances, and emergency situations. The proposed system also reduces the workload of healthcare professionals by automating routine health monitoring and maintaining continuous patient records. Since all physiological parameters are stored digitally, doctors can analyze historical health trends and make better clinical decisions. Overall, the system combines embedded systems, biomedical sensors, wireless communication, cloud computing, and IoT technologies into a single integrated healthcare solution that supports remote patient monitoring and early diagnosis.

3.2 System Components

The main components used in the proposed IoT-Based Remote Patient Vital Monitoring System are:

- ESP32 Development Board – Acts as the main controller and processes sensor data.
- MAX30102 Pulse Rate and SpO₂ Sensor – Measures heart rate and blood oxygen saturation (SpO₂).
- AD8232 ECG Sensor – Monitors the electrical activity of the heart (ECG).
- Temperature Sensor (MLX90614/DS18B20) – Measures the patient's body temperature.
- Blood Pressure Sensor – Measures systolic and diastolic blood pressure.
- Wi-Fi Module (Built into ESP32) – Transmits patient data to the IoT cloud.
- IoT Cloud Platform – Stores and displays patient health data for remote monitoring.
- Arduino IDE – Used to program the ESP32 microcontroller.
- Breadboard – Used for assembling and testing the circuit.
- Connecting Wires (Jumper Wires) – Used to connect all hardware components.

- Power Supply (USB/5V Adapter) – Provides power to the system.

3.3 Working of the Proposed System

The proposed IoT-Based Remote Patient Vital Monitoring System continuously monitors a patient's vital health parameters using biomedical sensors integrated with an ESP32 Development Board. When the system is powered on, the ESP32 initializes all the connected sensors, including the MAX30102 Pulse Rate and SpO₂ Sensor, Temperature Sensor, Blood Pressure Sensor, and AD8232 ECG Sensor. Each sensor collects the corresponding physiological parameter from the patient. The MAX30102 measures the heart rate and blood oxygen saturation (SpO₂), the temperature sensor measures body temperature, the blood pressure sensor records systolic and diastolic pressure, and the AD8232 sensor captures the electrical activity of the heart (ECG). The ESP32 receives these sensor readings, processes the data, and converts them into meaningful health values.

After processing, the ESP32 uses its built-in Wi-Fi module to transmit the collected data to an IoT cloud platform in real time. The uploaded data are securely stored and displayed on a web dashboard or mobile application, allowing doctors and caregivers to monitor the patient's health remotely from any location with internet access. The system continuously updates the patient's vital parameters, making it possible to observe changes in health conditions and identify abnormal readings at an early stage. This real-time monitoring improves the quality of healthcare, reduces the need for frequent hospital visits, and provides a cost-effective solution for home healthcare, telemedicine, elderly care, and remote patient monitoring applications.

3.4 Block Diagram

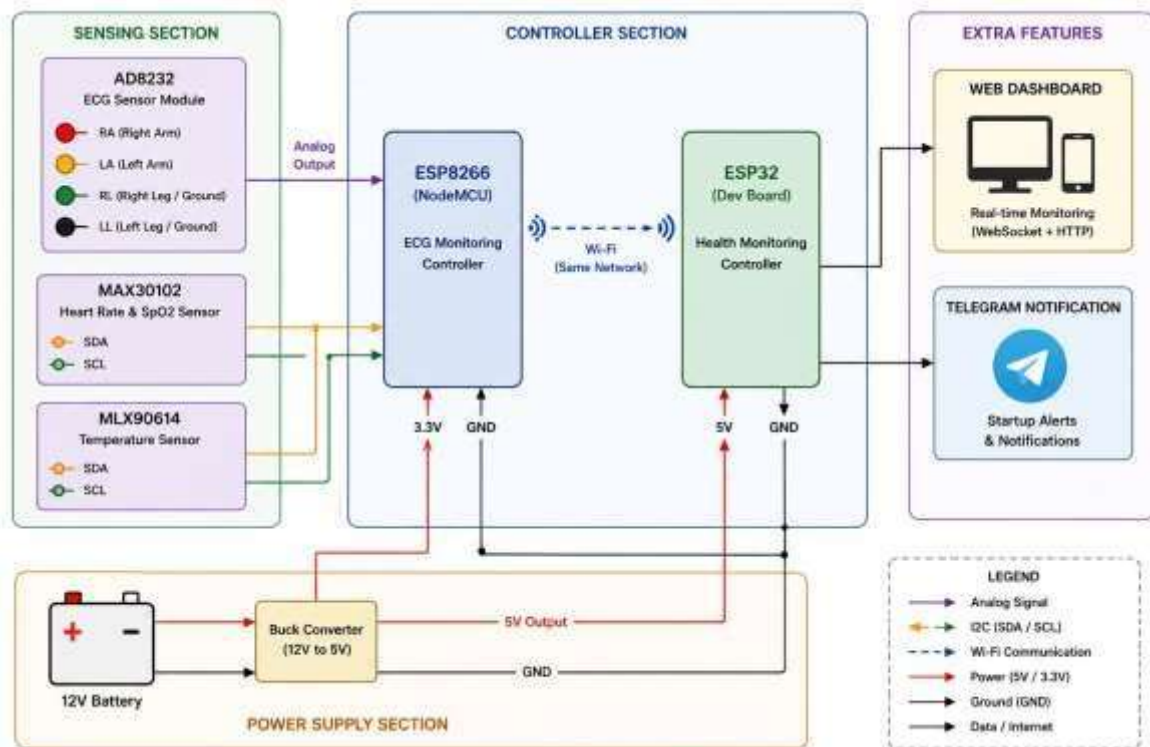


Fig:3.1: Block Diagram of Proposed System

3.5 Features of the Proposed System

- Real-Time Monitoring
- Remote Healthcare
- Multi-Parameter Monitoring
- Portable Design
- IoT Connectivity
- Continuous Data Storage
- User-Friendly Interface
- Low Power Consumption
- Secure Data Transmission
- Cost-Effective Solution

3.6 Advantages of the Proposed System

- Early Detection of Diseases
- Reduced Hospital Visits
- Improved Healthcare Accessibility
- Real-Time Medical Assistance
- Continuous Health Monitoring
- Cost-Effective Healthcare
- Better Patient Comfort

- Reduced Workload for Healthcare Professionals
- Scalability
- Future-Ready Platform

CHAPTER 4 HARDWARE AND SOFTWARE DESCRIPTION

4.1 Hardware Components

4.1.1 ESP32 Development Board

The ESP32 Development Board is a powerful, low-cost, and energy-efficient microcontroller board developed by Espressif Systems. It is designed for Internet of Things (IoT), embedded systems, and wireless communication applications. The ESP32 integrates a high-performance dual-core processor, built-in Wi-Fi and Bluetooth (BLE) connectivity, multiple General Purpose Input/Output (GPIO) pins, analog-to-digital converters (ADC), digital-to-analog converters (DAC), Pulse Width Modulation (PWM), SPI, I²C, UART, and other communication interfaces into a single compact chip. These features make it an ideal platform for developing smart devices that require wireless communication and real-time data processing. The ESP32 supports programming through the Arduino IDE, ESP-IDF, PlatformIO, and MicroPython, making it suitable for both beginners and advanced developers. It offers sufficient processing power and memory to handle multiple tasks simultaneously while consuming relatively low power, making it suitable for batteryoperated devices. The board also includes flash memory for program storage, voltage regulation circuitry, USB-to-Serial interface, and reset/boot buttons for easy programming and debugging.



Fig:4.1: ESP32 Development Board

4.1.2 MLX90614 Infrared Temperature Sensor

The MLX90614 Infrared Temperature Sensor is a high-precision, non-contact temperature sensor developed by Melexis. It measures the temperature of an object by detecting the infrared (IR) radiation naturally emitted from its surface, eliminating the need for direct physical contact. This feature makes it ideal for applications where contact-based measurement is difficult, unsafe, or unhygienic, such as medical monitoring, industrial automation, food processing, and IoT-based healthcare systems. The sensor integrates an infrared thermopile detector, a lownoise amplifier, a 17-bit Analog-to-Digital Converter (ADC), and a Digital Signal Processor (DSP) in a compact TO-39 metal package, providing highly accurate and stable temperature readings.

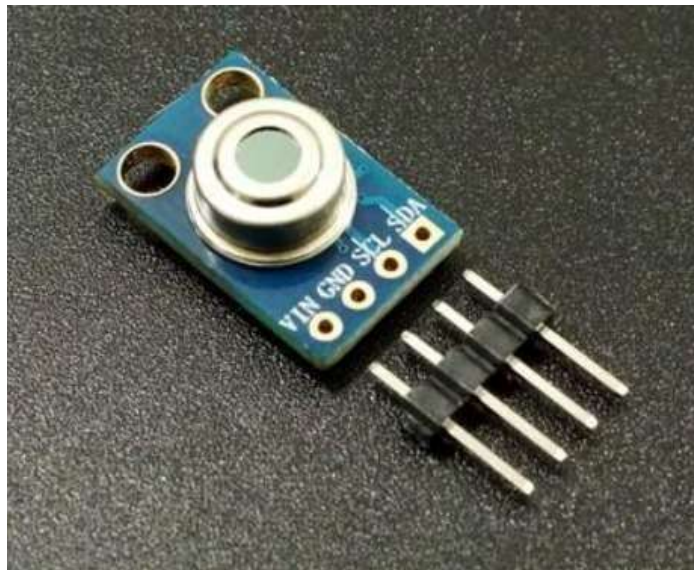


Fig:4.2: MLX90614 Infrared Temperature Sensor

4.1.3 AD8232 ECG Heart Rate Sensor Module

The AD8232 ECG Heart Rate Sensor Module is a compact and low-power biomedical sensor designed to measure the electrical activity of the human heart. Developed by Analog Devices, the AD8232 is an integrated signal conditioning block specifically designed for portable Electrocardiogram (ECG) and heart rate monitoring applications. It amplifies, filters, and processes weak electrical signals generated by the heart, producing a clean analog output that can be easily read by microcontrollers such as the ESP32, Arduino, and Raspberry Pi. The module uses

three disposable ECG electrodes placed on the human body to capture cardiac signals, making it suitable for real-time heart monitoring systems. The AD8232 incorporates an instrumentation amplifier, operational amplifier, high-pass and lowpass filters, and a right-leg drive amplifier within a single chip. These components work together to remove noise caused by body movement, muscle contractions, and electrical interference, resulting in accurate ECG waveforms. The module provides an analog output that can be converted into digital values using the Analog-to-Digital Converter (ADC) of the microcontroller. Its low power consumption and compact design make it highly suitable for wearable healthcare devices, remote patient monitoring systems, fitness trackers, and biomedical research.

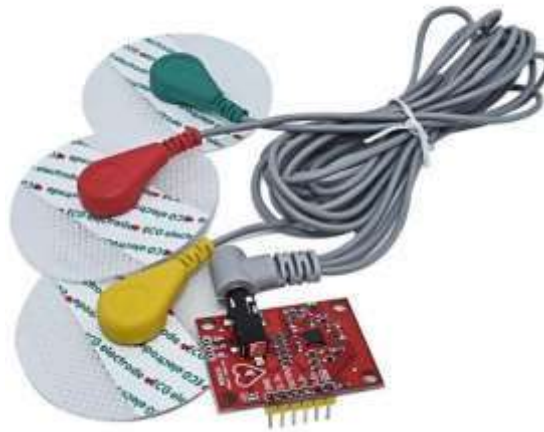


Fig:4.3: AD8232 ECG Heart Rate Sensor Module

4.1.4 MAX30102 Pulse Oximeter Sensor

The MAX30102 Pulse Oximeter Sensor is a compact, low-power biomedical sensor developed by Analog Devices (formerly Maxim Integrated) for measuring heart rate (Pulse Rate) and blood oxygen saturation (SpO_2). It combines two LEDs (red and infrared), a photodetector, optical components, and low-noise analog signal processing circuitry into a single miniature module. The sensor operates based on the principle of photoplethysmography (PPG), where red and infrared light are transmitted through the fingertip, and the reflected light is analyzed to determine the blood oxygen level and pulse rate. Due to its high accuracy and low power consumption, the MAX30102 is widely used in wearable healthcare devices, fitness trackers, smartwatches, and IoT-based patient monitoring systems. The MAX30102 communicates with microcontrollers such as the ESP32, Arduino, and Raspberry Pi.

using the I²C communication protocol. It contains an integrated Analog-to-Digital Converter (ADC), programmable LED current control, ambient light cancellation, and advanced digital filtering, which significantly improves measurement accuracy by minimizing interference from external light and motion. The sensor continuously measures pulse rate and oxygen saturation in real time, making it highly suitable for remote healthcare applications. Its compact size, low power requirements, and easy interfacing make it an excellent choice for portable medical devices.



Fig:4.4: MAX30102 Pulse Oximeter Sensor

4.1.5 Breadboard and Connecting Wires

A breadboard is used to build and test the circuit without soldering the components. It allows easy connection and modification of circuit components during the development phase. Jumper wires are used to connect the different components on the breadboard. These wires help establish electrical connections between the Arduino board, sensors, resistors, LEDs, and relay module. Using a breadboard simplifies the process of assembling and testing the circuit during the prototype stage.

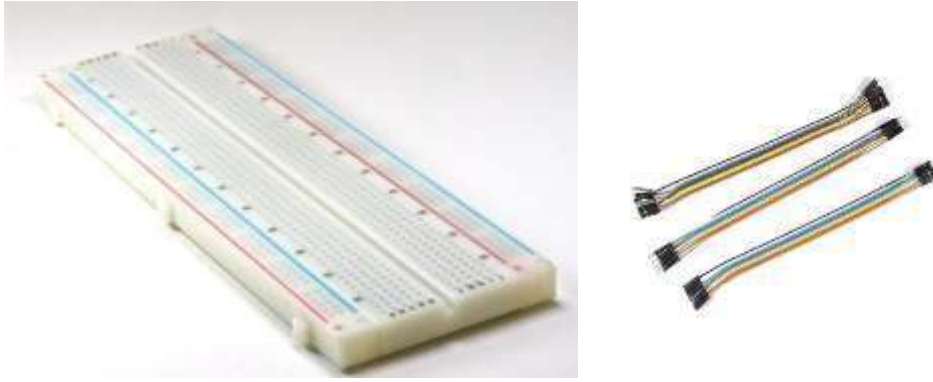


Fig:4.5: Breadboard and Connecting Wires

Jumper wires are used to connect the different components on the breadboard. These wires help establish electrical connections between the Arduino board, sensors, resistors, LEDs, and relay module. Using a breadboard simplifies the process of assembling and testing the circuit during the prototype stage.

4.2 Software Description

4.2.1 Arduino IDE

Arduino IDE is the software used to write, compile, and upload the program to the Arduino board. It provides a simple interface for coding and debugging. It supports Embedded C programming and includes built-in libraries for easy implementation. The serial monitor feature is used to view sensor values and debug the system.

4.2.2 IoT Monitoring Dashboard

The IoT monitoring dashboard is used to display the patient's vital health parameters received from the ESP32 in real time. The dashboard provides a simple interface for viewing heart rate, SpO₂, body temperature, blood pressure, and ECG readings in an organized manner. The data transmitted through Wi-Fi are displayed using numerical values, graphs, or indicators, making it easy for healthcare professionals and caregivers to observe the patient's health condition remotely. The dashboard can also maintain historical data, which helps in analyzing changes in vital parameters over time. If abnormal values are detected, an alert can be generated to notify the concerned person. Thus, the dashboard provides an effective interface between the IoT monitoring device and the healthcare professional, enabling realtime and remote patient monitoring.

CHAPTER 5 SYSTEM DESIGN

5.1 Circuit Diagram

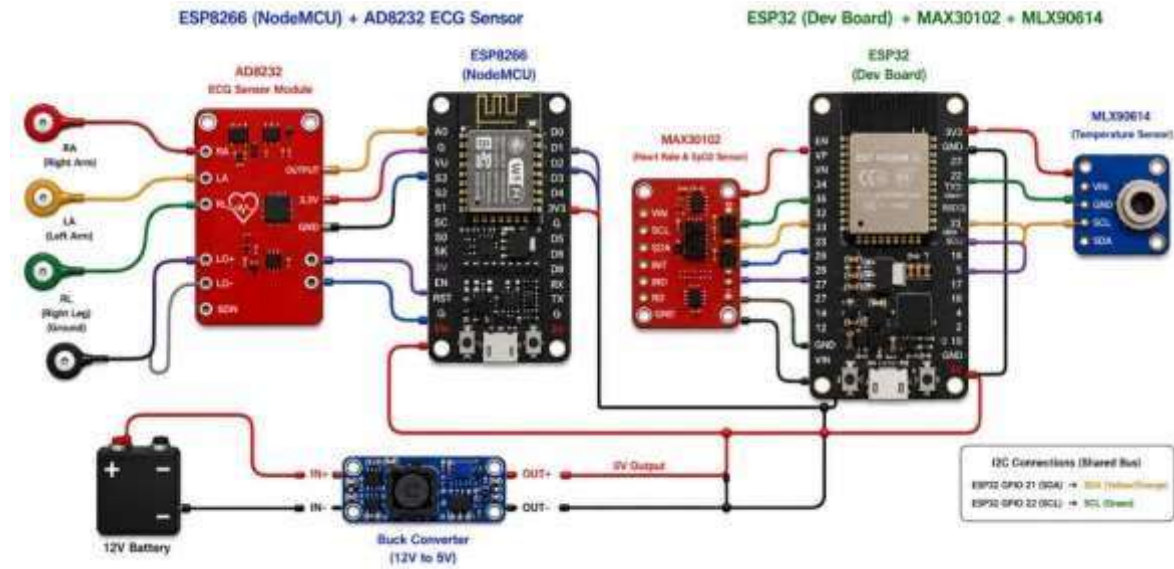


Fig. 5.1: IoT-Based Remote Patient Vital Monitoring System.

5.2 Working Principle

The proposed IoT-Based Remote Patient Vital Monitoring System operates by continuously measuring the patient's vital physiological parameters and transmitting the collected data to an IoT cloud platform for remote monitoring. Initially, the ESP32 Development Board initializes all the connected biomedical sensors, including the MAX30102 Pulse Rate and SpO₂ Sensor, Temperature Sensor, Blood Pressure Sensor, and AD8232 ECG Sensor. Once initialized, each sensor begins collecting real-time physiological data from the patient. The MAX30102 sensor measures heart rate and blood oxygen saturation by detecting changes in light absorption through the fingertip, while the temperature sensor measures body temperature. The blood pressure sensor records systolic and diastolic blood pressure values, and the AD8232 sensor captures the electrical activity of the heart to generate an ECG waveform. The ESP32 receives the sensor data, performs basic signal processing and validation, and converts the raw values into meaningful physiological parameters. Using its built-in Wi-Fi module, the processed data are transmitted securely to the IoT cloud platform. The cloud stores the patient's health records and displays the measurements on a web dashboard or mobile application in real time. Healthcare professionals can access the patient's

vital signs remotely, monitor health conditions continuously, and provide timely medical assistance if abnormal values are detected. This continuous monitoring process repeats automatically, ensuring uninterrupted observation of the patient's health status.

5.3 Algorithm

Step 1: Start the system.

Step 2: Initialize the ESP32 Development Board.

Step 3: Initialize the MAX30102, Temperature Sensor, Blood Pressure Sensor, and AD8232 ECG Sensor.

Step 4: Connect the ESP32 to the available Wi-Fi network.

Step 5: Read the patient's heart rate using the MAX30102 sensor.

Step 6: Measure the patient's blood oxygen saturation (SpO_2).

Step 7: Measure the patient's body temperature.

Step 8: Measure the patient's blood pressure.

Step 9: Acquire the ECG signal from the AD8232 sensor.

Step 10: Process and validate all sensor readings.

Step 11: Display the measured values on the Serial Monitor or local display.

Step 12: Upload the processed data to the IoT cloud platform.

Step 13: Update the cloud dashboard with the latest patient health information.

Step 14: If any parameter exceeds the normal threshold, generate an alert notification.

Step 15: Repeat Steps 5–14 continuously for real-time monitoring.

Step 16: Stop the system.

5.4 Flowchart

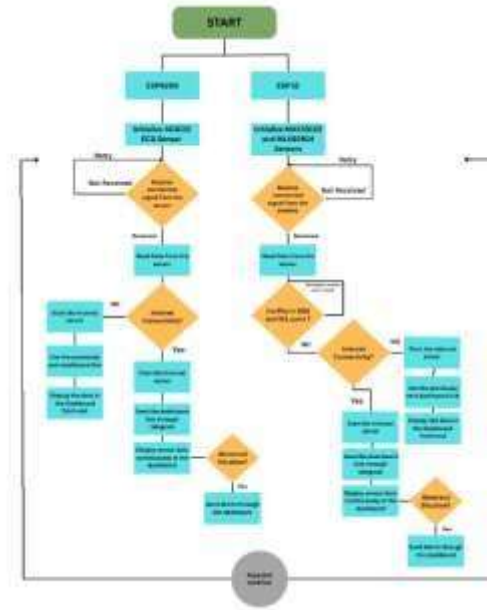


Fig:5.2: Flowchart of the Proposed System

The flowchart begins by initializing the ESP32 microcontroller and all biomedical sensors. The ESP32 then establishes a Wi-Fi connection and starts collecting physiological parameters, including heart rate, SpO₂, body temperature, blood pressure, and ECG signals. The collected data are processed and validated before being uploaded to the IoT cloud platform. The cloud dashboard displays the patient's health information in real time. If any measured parameter exceeds the predefined normal range, the system automatically generates an alert notification to healthcare professionals or caregivers. Otherwise, the monitoring process continues continuously, ensuring uninterrupted remote patient health monitoring.

CHAPTER 6

SYSTEM IMPLEMENTATION AND RESULTS

6.1 Implementation

The implementation of the IoT-Based Remote Patient Vital Monitoring System involved the integration of biomedical sensors with the ESP32 Development Board to monitor a patient's vital parameters in real time. The system was developed using

hardware components such as the ESP32 microcontroller, MAX30102 Pulse Rate and SpO₂ Sensor, Temperature Sensor, Blood Pressure Sensor, AD8232 ECG Sensor, breadboard, connecting wires, and a regulated power supply. These components were interconnected according to the circuit design and programmed using the Arduino IDE.

Initially, the ESP32 was configured to establish communication with all connected sensors through I²C, analog, and digital interfaces. Embedded C/C++ programming was used to initialize the sensors, read the physiological parameters, process the collected data, and transmit the readings to the IoT cloud platform using the ESP32's built-in Wi-Fi module. The cloud dashboard displayed the patient's heart rate, blood oxygen saturation, body temperature, blood pressure, and ECG waveform in real time. The developed prototype successfully demonstrated continuous patient monitoring and remote healthcare accessibility.

6.2 Experimental Setup

The experimental setup was designed to evaluate the performance and reliability of the proposed monitoring system under normal operating conditions.

The setup consisted of:

- ESP32 Development Board
- MAX30102 Pulse Rate and SpO₂ Sensor
- Body Temperature Sensor
- Blood Pressure Sensor
- AD8232 ECG Sensor
- Breadboard and Connecting Wires
- USB Power Supply
- Wi-Fi Internet Connection
- Laptop/Smartphone with IoT Dashboard

During testing, the sensors were connected to the ESP32 through the breadboard. The patient's fingertip was placed on the MAX30102 sensor to measure heart rate and SpO₂, while the temperature sensor measured body temperature.

The blood pressure sensor recorded blood pressure values, and the AD8232 ECG sensor was connected using electrodes to capture ECG signals. The ESP32 processed the sensor data and transmitted it to the cloud platform through Wi-Fi. The measured values were viewed in real time using a mobile phone or computer connected to the Internet.

6.3 Testing Methodology

The developed system was tested to verify the accuracy and reliability of measuring different physiological parameters. Each sensor was tested individually and then as part of the integrated system. Multiple readings were recorded under normal conditions to evaluate the consistency of the measurements.

The testing procedure involved the following steps:

1. Power ON the ESP32 development board.
2. Connect the ESP32 to the Wi-Fi network.
3. Initialize all biomedical sensors.
4. Place the sensors correctly on the patient.
5. Measure heart rate, SpO₂, body temperature, blood pressure, and ECG.
6. Process the sensor data using the ESP32.
7. Upload the processed data to the IoT cloud platform.
8. Display the measured values on the cloud dashboard.
9. Compare the obtained values with expected normal ranges to verify system performance.

The test was repeated several times to ensure reliable and consistent operation.

6.4 Test Cases and Results

The proposed IoT-Based Remote Patient Vital Monitoring System was tested by checking each sensor and the overall system under normal operating conditions. Initially, the ESP32 was powered on and connected to the Wi-Fi network to verify proper initialization and communication. The MAX30102 sensor was tested by placing a finger on the sensor to measure heart rate and SpO₂, and the readings were successfully obtained. The temperature sensor was tested to measure the patient's body temperature, while the blood pressure sensor was used to obtain

systolic and diastolic pressure values. The AD8232 ECG sensor was tested by attaching the electrodes to the appropriate body positions, and the ECG waveform was successfully obtained. After collecting the sensor readings, the ESP32 processed the data and transmitted them to the IoT cloud platform. The dashboard was checked to confirm that the patient's vital parameters were displayed correctly and updated in real time.

The system was also tested for continuous monitoring and remote data transmission. During testing, the ESP32 maintained a stable Wi-Fi connection and successfully uploaded the collected health data to the cloud platform. Different sensor readings were observed to verify that the system could identify changes in the patient's vital parameters. The system was able to monitor heart rate, SpO₂, body temperature, blood pressure, and ECG simultaneously. When abnormal values were detected, the system was designed to generate an alert for further attention. The test results demonstrated that the proposed system can provide reliable real-time monitoring and remote access to patient health information. Overall, the test cases confirmed the proper operation of the sensors, ESP32 controller, wireless communication, and IoT dashboard, making the system suitable for remote and continuous patient monitoring.

6.5 Result Analysis

The experimental results demonstrate that the proposed IoT-Based Remote Patient Vital Monitoring System successfully monitors multiple vital health parameters, including heart rate, blood oxygen saturation (SpO₂), body temperature, blood pressure, and ECG signals. The ESP32 efficiently processed the sensor data and transmitted the readings to the IoT cloud platform with minimal delay. Real-time monitoring enabled healthcare professionals to access patient information remotely using smartphones or computers. The sensor readings remained stable during repeated tests, indicating reliable system performance. The integration of multiple sensors into a single portable device reduced the need for separate medical equipment and improved the convenience of patient monitoring. Overall, the system achieved its objectives by providing accurate, continuous, and cost-effective remote healthcare monitoring suitable for home care, telemedicine, and rural healthcare applications.

6.6 Prototype Model

The prototype model of the IoT-Based Remote Patient Vital Monitoring System consists of an ESP32 Development Board connected to a MAX30102 Pulse Rate and SpO₂ Sensor, Temperature Sensor, Blood Pressure Sensor, and AD8232 ECG Sensor using a breadboard and jumper wires. The ESP32 acts as the central controller, collecting physiological data from all sensors, processing the information, and transmitting it to the IoT cloud platform through its built-in Wi-Fi module. The prototype is compact, portable, and easy to operate, making it suitable for real-time patient monitoring in homes, hospitals, rural health centers, and emergency medical services. The developed model successfully demonstrates the integration of embedded systems, biomedical sensors, and IoT technology to provide continuous remote healthcare monitoring.



Fig.6.1: Dashboard of the IoT-Based Remote Patient Vital Monitoring System



Fig.6.2: Dashboard of the IoT-Based Remote Patient Vital Monitoring System

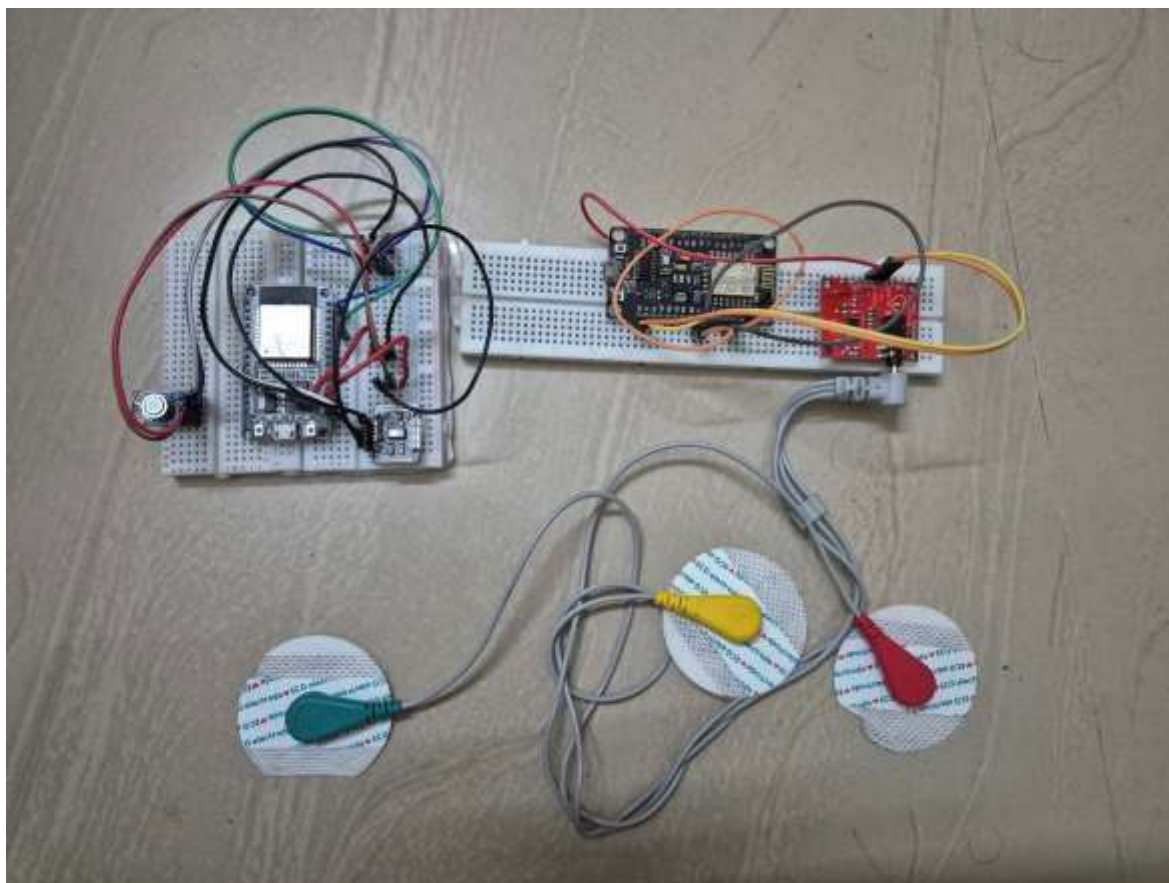


Fig:6.3: Prototype model for IoT-Based Remote Patient Vital Monitoring System

CHAPTER 7 CONCLUSION

The IoT-Based Remote Patient Vital Monitoring System was successfully designed and implemented to provide a portable, reliable, and cost-effective solution for continuous patient health monitoring. The system integrates multiple biomedical sensors, including a MAX30102 Pulse Rate and SpO₂ Sensor, Temperature Sensor, Blood Pressure Sensor, and AD8232 ECG Sensor, with an ESP32 Development Board to measure essential physiological parameters such as heart rate, blood oxygen saturation SpO₂, body temperature, blood pressure, and ECG signals. The collected data are processed by the ESP32 and transmitted to an IoT cloud platform through its built-in Wi-Fi module, enabling healthcare professionals to monitor patients remotely in real time. The developed system effectively demonstrates the advantages of combining embedded systems, IoT technology, and biomedical sensors to improve healthcare services. Continuous monitoring of vital signs allows

the early detection of abnormal health conditions, enabling timely medical intervention and reducing the risk of severe complications. The wireless transmission of patient data eliminates the need for frequent hospital visits and supports home-based healthcare, telemedicine, elderly care, and remote patient monitoring, particularly in rural and underserved areas where access to medical facilities is limited.

Experimental testing confirmed that the system accurately measured and transmitted patient vital parameters with reliable performance. The sensor readings were displayed in real time on the IoT platform, allowing doctors and caregivers to access patient information using smartphones, tablets, or computers from any location with internet connectivity. The integration of multiple sensors into a single compact device simplifies patient monitoring while reducing the need for separate medical equipment. The prototype also demonstrated stable operation, low power consumption, ease of use, and efficient wireless communication, making it suitable for practical healthcare applications. In conclusion, the proposed IoT-Based Remote Patient Vital Monitoring System successfully achieves the objectives of providing continuous, real-time, and remote monitoring of patient vital signs. The system offers an affordable and user-friendly healthcare solution that improves patient safety, enhances medical decision-making, and supports preventive healthcare. Future enhancements may include the integration of Artificial Intelligence (AI) for disease prediction, automatic emergency alert notifications, wearable sensor technology, GPS-based patient tracking, cloud-based medical record management, and advanced data analytics to further improve the efficiency and effectiveness of smart healthcare systems.

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